

Algebra II - Tutorial 5 - Date: 23/2/2026

1. Show that the group law on $H \rtimes_{\phi} K$ is associative.
2. Show that $S_n = A_n \rtimes \mathbb{Z}_2$
3. Show that $D_{2n} = D_n \times C_2$ (n odd)
4. $D_{2n} = C_n \rtimes C_2$ for a suitable homomorphism $\phi : C_2 \rightarrow \text{Aut}(C_n)$
5. Can we write $C_4 = C_2 \rtimes C_2$
6. Let x be a generator of C_4 . Let $\phi : C_4 \rightarrow \text{Aut}(C_3)$ be the unique homomorphism such that $\phi_x(h) = h^{-1}$. Show that the semidirect product $C_3 \rtimes_{\phi} C_4$ is a nonabelian group of order 12 which is isomorphic to neither A_4 nor D_6 . In fact this is the group $G = \langle x, y : x^4 = 1, y^3 = 1, xy = y^2x \rangle$.
7. Let p and q be primes such that $p < q$.
 - (a) Show that if p does not divide $q - 1$, then the only homomorphism from C_p to $\text{Aut}(C_q)$ is the trivial map (which sends every element to identity).
 - (b) Show that if p divides $q - 1$, then there is a nontrivial homomorphism $\phi : C_p \rightarrow \text{Aut}(C_q)$. Further show that if $\psi : C_p \rightarrow \text{Aut}(C_q)$ is any other nontrivial homomorphism, then the semidirect products $C_q \rtimes_{\phi} C_p$ and $C_q \rtimes_{\psi} C_p$ are isomorphic. Deduce that, if p divides $q - 1$, then up to isomorphism there is only one nonabelian group of order pq
8. Show that the quaternion group Q_8 is not a semidirect product of any of its two proper subgroups.
9. Determine whether $C_6 = C_3 \rtimes C_2$
10. Determine whether $S_3 = C_3 \rtimes C_2$
11. Show that $GL_n(\mathbb{R}) = SL_n(\mathbb{R}) \rtimes \text{diag}(a, 1, \dots, 1)$ where $a \in \mathbb{R} - \{0\}$ and .